

Brazilian E-Commerce Sales Analysis

SQL | Power BI | Excel | SQLite

Portfolio Project

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- **Date**

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1. Project Overview

Introduction

This project presents an end-to-end business analysis of the **Brazilian E-commerce Public Dataset (Olist)** using **SQL** and **Power BI**. The objective was to transform raw transactional data into meaningful business insights that support data-driven decision-making.

The analysis focuses on multiple business areas, including sales performance, customer behavior, operational efficiency, and financial performance. SQL was used to extract, clean, and analyze the data, while Power BI was used to create interactive dashboards that visualize the findings.

Throughout this project, **20 business questions** were answered, covering key performance indicators (KPIs), customer segmentation, revenue trends, delivery performance, seasonal purchasing behavior, and business opportunities.

The final outcome is a comprehensive business report supported by SQL queries, interactive Power BI dashboards, and actionable recommendations that demonstrate practical data analysis skills.

2. Dataset

Dataset Source

Dataset Name: Brazilian E-Commerce Public Dataset by Olist

Source: Kaggle

Database: SQLite

Dataset Description

The dataset contains real e-commerce transactions from Olist, one of the largest Brazilian online marketplaces. It includes information about customers, orders, products, payments, sellers, reviews, and delivery operations.

The data provides a comprehensive view of the entire customer journey, making it suitable for business intelligence, sales analysis, customer analytics, and operational performance evaluation.

Selected Tables

TABLE	DESCRIPTION
CUSTOMERS	Customer information including location and unique customer identifiers.
ORDERS	Order details, purchase dates, delivery dates, and order status.
ORDER ITEMS	Products purchased in each order, including prices and freight costs.
PRODUCTS	Product information and product categories.
Payments	Payment values and payment methods used for each order.
REVIEWS	Customer review scores and review information after order delivery.

3. Data Cleaning

Data Cleaning Process

Before performing the analysis, the dataset was reviewed and prepared to ensure data quality and consistency.

The following steps were completed:

- Verified the integrity of relationships between all tables.
- Checked for missing values in the selected tables.
- Reviewed duplicate records to ensure data accuracy.
- Confirmed appropriate data types for dates, numerical values, and categorical fields.
- Filtered the analysis to completed (delivered) orders where required to ensure accurate business metrics.
- Validated calculations before building the Power BI dashboards.

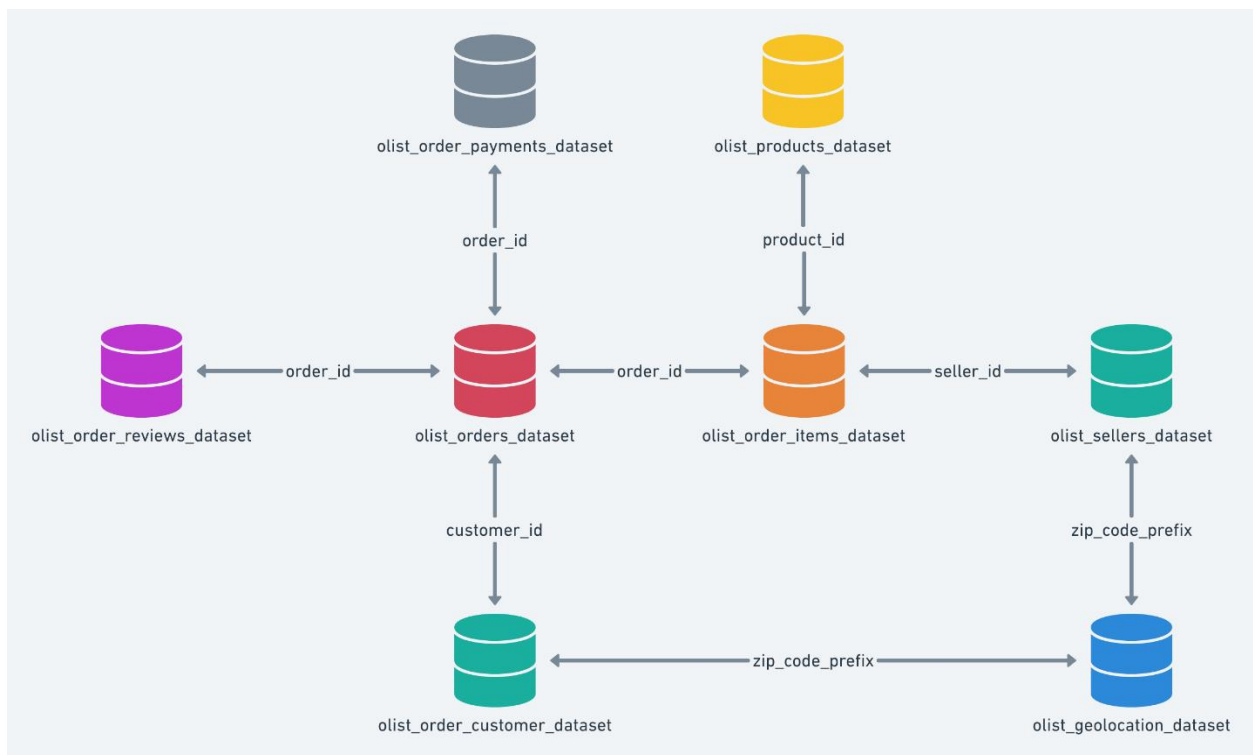
These preparation steps ensured that the analysis was based on reliable and consistent data.

4. Database Schema

This section presents the database structure used in the analysis, including the relationships between the main tables. Understanding these relationships ensures accurate data integration and supports reliable business analysis.

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the relationships between the main tables used in this project. These relationships enabled accurate data integration across customers, orders, products, payments, and reviews, providing a solid foundation for SQL analysis and Power BI reporting.



5. SQL Analysis

This section presents the SQL analysis performed to answer key business questions and extract valuable insights from the Brazilian E-commerce dataset.

Note:

- All SQL queries used in this project are available in the attached `Business_Analysis.sql` file

• 5.1 Sales Performance Analysis

This section analyzes the overall sales performance of the Brazilian e-commerce platform, including revenue generation, sales trends, product performance, and average order value. The analysis aims to identify key sales patterns and business opportunities.

1. Total Revenue Generated

Objective

Calculate the total revenue generated from all completed customer orders.

Business Insight

The platform generated **more than 15 million** in total revenue from completed orders, reflecting strong sales performance and a high level of business activity. This metric serves as a key indicator of the company's overall financial performance.

Analysis Result

	total_revenue
1	15422461.77

2. Monthly Sales Trend

Objective

Analyze monthly revenue trends to identify seasonality, growth patterns, and fluctuations in sales performance over time.

Business Insight

Monthly revenue showed noticeable fluctuations throughout the analyzed period, highlighting seasonal demand patterns and changes in customer purchasing behavior. Understanding these trends can support better sales forecasting and business planning.

Analysis Result

	order_month	total_revenue
1	2016-10	46566.71
2	2016-12	19.62
3	2017-01	127545.67
4	2017-02	271298.65
5	2017-03	414369.39
6	2017-04	390952.18
7	2017-05	567066.73
8	2017-06	490225.6
9	2017-07	566403.93
10	2017-08	646000.61
11	2017-09	701169.99
12	2017-10	751140.27
13	2017-11	1153528.05
14	2017-12	843199.17

3. Top Product Categories by Revenue

Objective

Identify the product categories that generate the highest revenue.

Business Insight

A relatively small number of product categories contributed the majority of total revenue. These high-performing categories represent valuable opportunities for targeted marketing campaigns, inventory optimization, and strategic business investment.

Analysis Result

	category	total_revenue
1	beleza_saude	1233131.72
2	relogios_presentes	1166176.98
3	cama_mesa_banho	1023434.76
4	esporte_lazer	954852.55
5	informatica_acessorios	888724.61
6	moveis_decoracao	711927.69
7	utilidades_domesticas	615628.69
8	cool_stuff	610204.1
9	automotivo	578966.65
10	brinquedos	471286.48
11	ferramentas_jardim	470495.28
12	bebes	400421.84
13	perfumaria	390144.65
14	telefonica	309860.23

4. Average Order Value

Objective

Calculate the average amount spent per customer order.

Business Insight

The average order value provides valuable insight into customer purchasing behavior and spending habits. Monitoring this KPI helps evaluate pricing strategies and measure the effectiveness of sales initiatives.

Analysis Result

average_order_value
159.86

5. Best-Selling Products

Objective

Identify the products with the highest sales volume.

Business Insight

The analysis revealed that a small number of products consistently achieved significantly higher sales volumes than others. These products represent the company's strongest performers and should be prioritized in inventory planning, promotional campaigns, and demand forecasting.

Analysis Result

	product_id	total_sales
1	aca2eb7d00eala7b8ebd4e68314663af	520
2	422879e10f46682990de24d770e7f83d	484
3	99a4788cb24856965c36a24e339b6058	477
4	389d119b48cf3043d311335e499d9c6b	390
5	368c6c730842d78016ad823897a372db	388
6	53759a2ecddad2bb87a079a1f1519f73	373
7	d1c427060a0f73f6b889a5c7c61f2ac4	332
8	53b36df67ebb7c41585e8d54d6772e08	321
9	154e7e31ebfa092203795c972e5804a6	274
10	3dd2a17168ec895c781a9191cle95ad7	272

• 5.2 Customer Analysis

This section focuses on customer purchasing behavior, spending patterns, customer loyalty, and geographical distribution. The objective is to better understand customer value and identify opportunities to improve retention and business growth.

6. Top Customers by Total Spending

Objective

Identify the customers who generated the highest total revenue.

Business Insight

The analysis revealed that a relatively small group of customers contributed a significant portion of the total revenue. These high-value customers represent an important business asset and can be targeted through loyalty programs, personalized offers, and customer retention strategies.

Analysis Result

	customer_unique_id	total_spent
1	0a0a92112bd4c708ca5fde585afaa872	13664.08
2	da122df9eeddfedc1dc1f5349a1a690c	7571.63
3	763c8b1c9c68a0229c42c9fc6f662b93	7274.88
4	dc4802a71eae9be1dd28f5d788ceb526	6929.31
5	459bef486812aa25204be022145caa62	6922.21
6	ff4159b92c40ebe40454e3e6a7c35ed6	6726.66
7	4007669dec559734d6f53e029e360987	6081.54
8	eebb5dda148d3893cdaf5b5ca3040ccb	4764.34
9	48elac109decbb87765a3eade6854098	4681.78
10	c8460e4251689ba205045f3ea17884a1	4655.91

7. Average Spending per Customer

Objective

Calculate the average amount spent by each customer.

Business Insight

The average spending per customer provides an estimate of the revenue generated by a typical customer. This metric helps evaluate customer value, purchasing behavior, and overall revenue contribution across the customer base.

Analysis Result

	average_spent
1	165.1988

8. Repeat Customers Spending Analysis

Objective

Compare the average spending of repeat customers with one-time customers.

Business Insight

The analysis showed that repeat customers spend more on average than one-time customers. This finding highlights the importance of customer retention strategies, as encouraging repeat purchases can significantly increase long-term revenue.

Analysis Result

	customer_type	avg_spending
1	One-time Customer	160.76
2	Repeat Customer	308.59

9. Top Cities and States by Number of Orders

Objective

Identify the cities and states with the highest order volumes.

Business Insight

Certain cities and states generated substantially more orders than others, indicating regions with stronger customer demand. These locations present valuable opportunities for focused marketing campaigns, logistics optimization, and business expansion.

Analysis Result

	city	state	total_orders
1	sao paulo	SP	15045
2	rio de janeiro	RJ	6601
3	belo horizonte	MG	2697
4	brasilia	DF	2071
5	curitiba	PR	1489
6	campinas	SP	1406
7	porto alegre	RS	1342
8	salvador	BA	1188
9	guarulhos	SP	1144
10	sao bernardo do campo	SP	911
11	niteroi	RJ	825
12	santo andre	SP	777

10.Repeat Customer Percentage

Objective

Measure the percentage of customers who placed more than one order.

Business Insight

The analysis revealed that the percentage of repeat customers is relatively low, suggesting opportunities to strengthen customer loyalty and improve retention. Implementing loyalty programs, personalized recommendations, and targeted promotions may help encourage repeat purchases and increase customer lifetime value.

Analysis Result

	repeat_customer_percentage
1	3.0

5.3 Delivery & Operations Analysis

This section evaluates delivery performance, shipping efficiency, and operational effectiveness. It also examines how delivery performance influences customer satisfaction and overall order fulfillment.

11.Average Delivery Time

Objective

Calculate the average time required to deliver customer orders.

Business Insight

The average delivery time reflects the efficiency of shipping and fulfillment operations. Monitoring this metric helps evaluate logistics performance and identify opportunities to improve customer experience.

Analysis Result

	avg_delivery_time
1	12.56

12.Percentage of Delayed Deliveries

Objective

Measure the percentage of orders delivered later than the estimated delivery date.

Business Insight

The analysis measures the proportion of delayed deliveries, providing valuable insight into shipping performance and operational efficiency. Reducing delivery delays can significantly improve customer satisfaction.

Analysis Result

	delayed_delivery_percentage
1	8.11

13.Does Delivery Delay Affect Customer Reviews?

Objective

Analyze the relationship between delivery delays and customer review scores.

Business Insight

The analysis showed that delayed deliveries generally received lower review scores than orders delivered on time, indicating that delivery performance has a direct impact on customer satisfaction and the overall shopping experience.

Analysis Result

	delivery_status	avg_review_score
1	Delayed	2.57
2	On-Time	4.29

14. Most Common Order Statuses

Objective

Identify the distribution of order statuses across all customer orders.

Business Insight

Most orders were successfully delivered, while canceled and incomplete orders represented only a small proportion of total orders. This indicates an efficient order fulfillment process and reliable operational performance.

Analysis Result

	order_status	total_orders	percentage
1	delivered	96478	97.02
2	shipped	1107	1.11
3	canceled	625	0.63
4	unavailable	609	0.61
5	invoiced	314	0.32
6	processing	301	0.3
7	created	5	0.01
8	approved	2	0.0

5.4 Financial & Business Insights

This section explores financial performance, pricing behavior, shipping costs, and profitability to identify business opportunities and support strategic decision-making.

15.Relationship Between Freight Cost and Product Price

Objective

Analyze the relationship between product prices and freight costs.

Business Insight

The analysis revealed a positive relationship between product price and average freight cost, suggesting that higher-value products generally incur higher shipping expenses due to factors such as size, weight, or handling requirements.

Analysis Result

	price_category	avg_freight_cost
1	Expensive	155.44
2	Medium	88.78
3	Cheap	19.93

16.Do Expensive Products Receive Better Reviews?

Objective

Evaluate whether product price influences customer review scores.

Business Insight

Customer review scores varied across different product price categories, indicating that higher-priced products do not necessarily receive better ratings. Customer satisfaction is influenced by multiple factors beyond price alone.

Analysis Result

	price_category	avg_rev
1	Expensive	4.8
2	Cheap	4.0326
3	Medium	3.8395

17.Product Categories with Highest Freight Cost

Objective

Identify product categories with the highest average freight costs.

Business Insight

Certain product categories exhibited considerably higher average freight costs than others, likely due to differences in product dimensions, weight, or shipping requirements. These findings can support logistics optimization and pricing strategies.

Analysis Result

	category	avg_freight_cost
1	pcs	48.45
2	eletrodomesticos_2	44.54
3	moveis_colchao_e_estofado	42.91
4	moveis_cozinha_area_de_servico_jant...	42.7
5	moveis_quarto	42.5
6	moveis_escritorio	40.55
7	portateis_casa_forno_e_cafe	36.16
8	moveis_sala	35.72
9	sinalizacao_e_seguranca	32.7
10	industria_comercio_e_negocios	29.42
11	malas_acessorios	27.88
12	agro_industria_e_comercio	27.56
13	instrumentos_musicais	27.41
14	construcao_ferramentas_iluminacao	24.95

18.Most Profitable Months

Objective

Identify the months that generated the highest revenue.

Business Insight

Revenue varied noticeably throughout the year, with several months outperforming others. These periods may reflect seasonal demand, promotional campaigns, or consumer purchasing trends, providing valuable input for future sales planning.

Analysis Result

	order_month	total_revenue
1	2017-11	1153528.05
2	2018-04	1132933.95
3	2018-05	1128836.69
4	2018-03	1120678.0
5	2018-01	1078606.86
6	2018-07	1027903.86
7	2018-06	1012090.68
8	2018-08	985414.28
9	2018-02	966510.88
10	2017-12	843199.17
11	2017-10	751140.27
12	2017-09	701169.99
13	2017-08	646000.61
14	2017-05	567066.73

5.5 Advanced Insights

This section presents advanced analytical findings by combining customer behavior and seasonal purchasing patterns to support data-driven business strategies.

19.Customer Segmentation Based on Spending

Objective

Segment customers into spending groups based on their total purchase value.

Business Insight

Customers were classified into different spending segments according to their total expenditure. This segmentation helps identify high-value customers, improve targeted marketing campaigns, and support more effective customer retention strategies.

Analysis Result

	customer_unique_id	total_spent	customer_segment
1	0a0a92112b...	13664.08	High Value
2	dal22df9ee...	7571.63	Medium ...
3	763c8b1c9c...	7274.88	Medium ...
4	dc4802a71e...	6929.31	Medium ...
5	459bef4868...	6922.21	Medium ...
6	ff4159b92c...	6726.66	Medium ...
7	4007669dec...	6081.54	Medium ...
8	eebb5dda14...	4764.34	Medium ...
9	48elac109d...	4681.78	Medium ...
10	c8460e4251...	4655.91	Medium ...
11	edde2314c6...	4513.32	Low Value
12	a229eba70e...	4445.5	Low Value
13	edf81elf30...	4194.76	Low Value

20. Seasonal Purchasing Behavior

Objective

Analyze order volume across different months to identify seasonal purchasing patterns.

Business Insight

The analysis revealed noticeable differences in monthly order volumes, indicating clear seasonal purchasing behavior. Recognizing these demand patterns enables better inventory management, marketing planning, and resource allocation during peak periods.

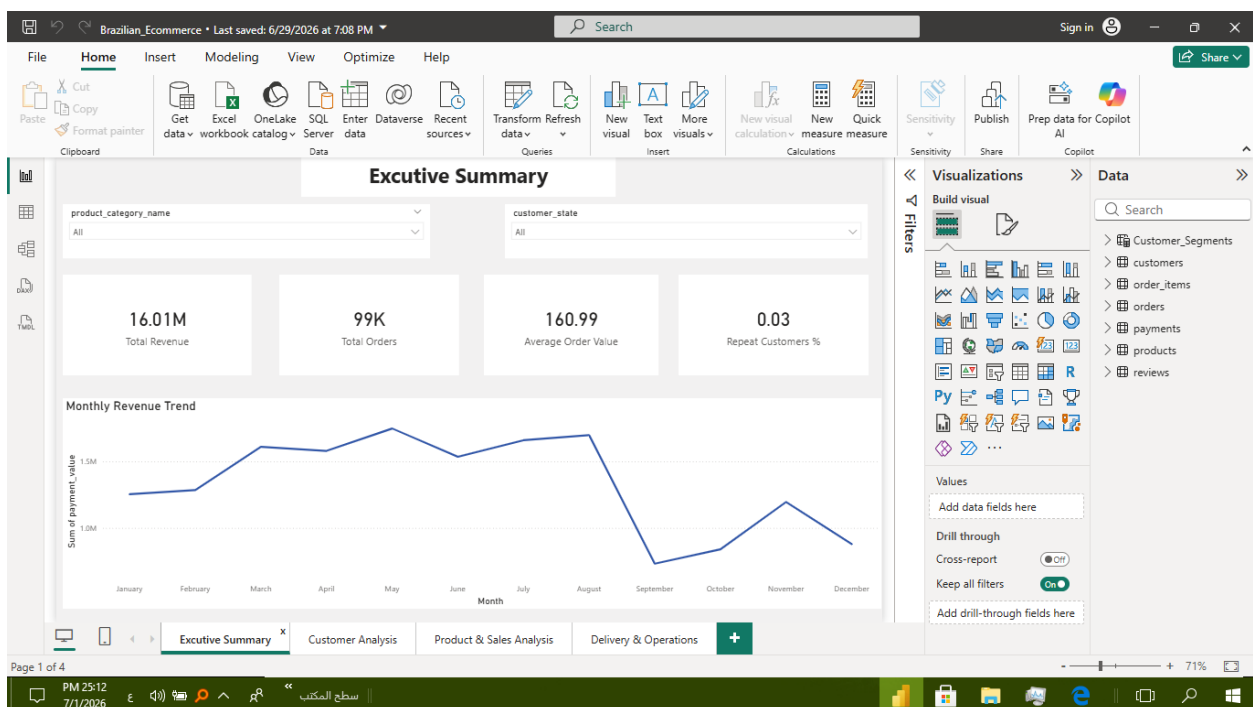
Analysis Result

	order_month	total_orders
1	08	10544
2	05	10295
3	07	10031
4	03	9549
5	06	9234
6	04	9101
7	02	8208
8	01	7819
9	11	7289
10	12	5514
11	10	4743
12	09	4151

6. Power BI Dashboard

The Power BI dashboard was developed to transform the SQL findings into interactive visualizations, enabling users to explore sales performance, customer behavior, operational efficiency, and business trends through dynamic reports.

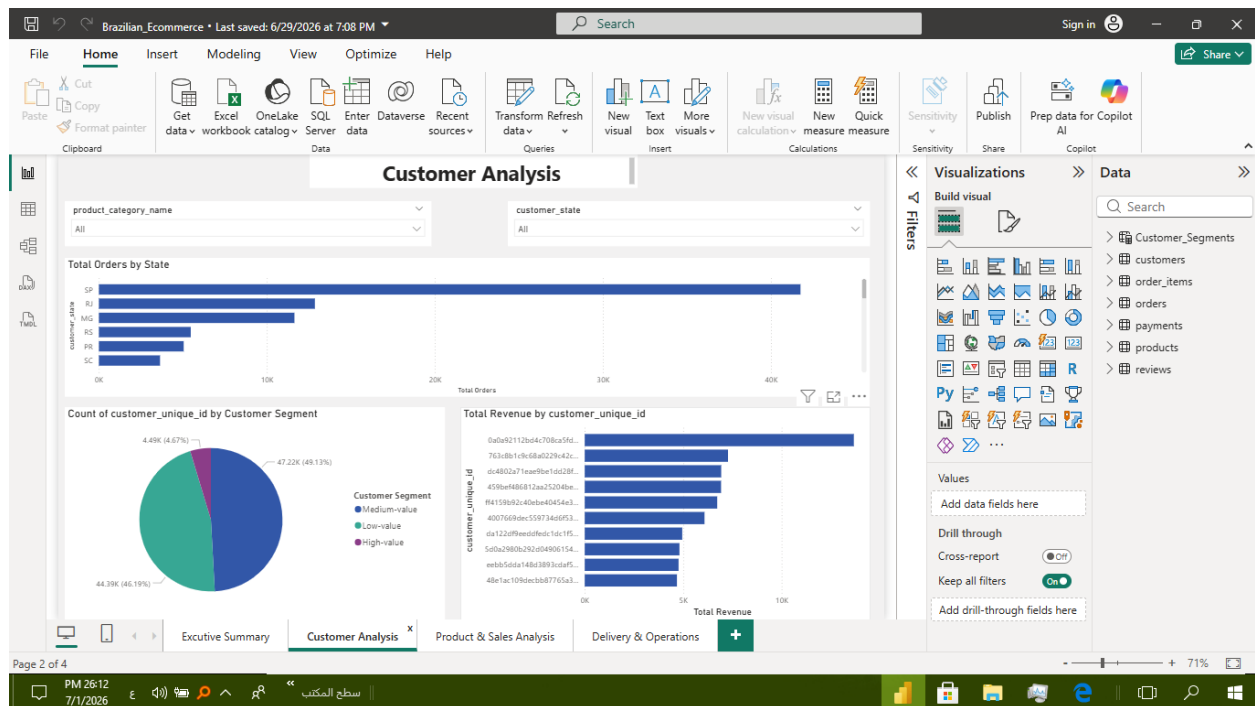
Executive Overview



Description

This dashboard provides a high-level overview of the company's overall performance through key performance indicators (KPIs), including total revenue, total orders, average order value, and repeat customer percentage. It also presents the monthly sales trend, enabling users to quickly evaluate overall business performance over time.

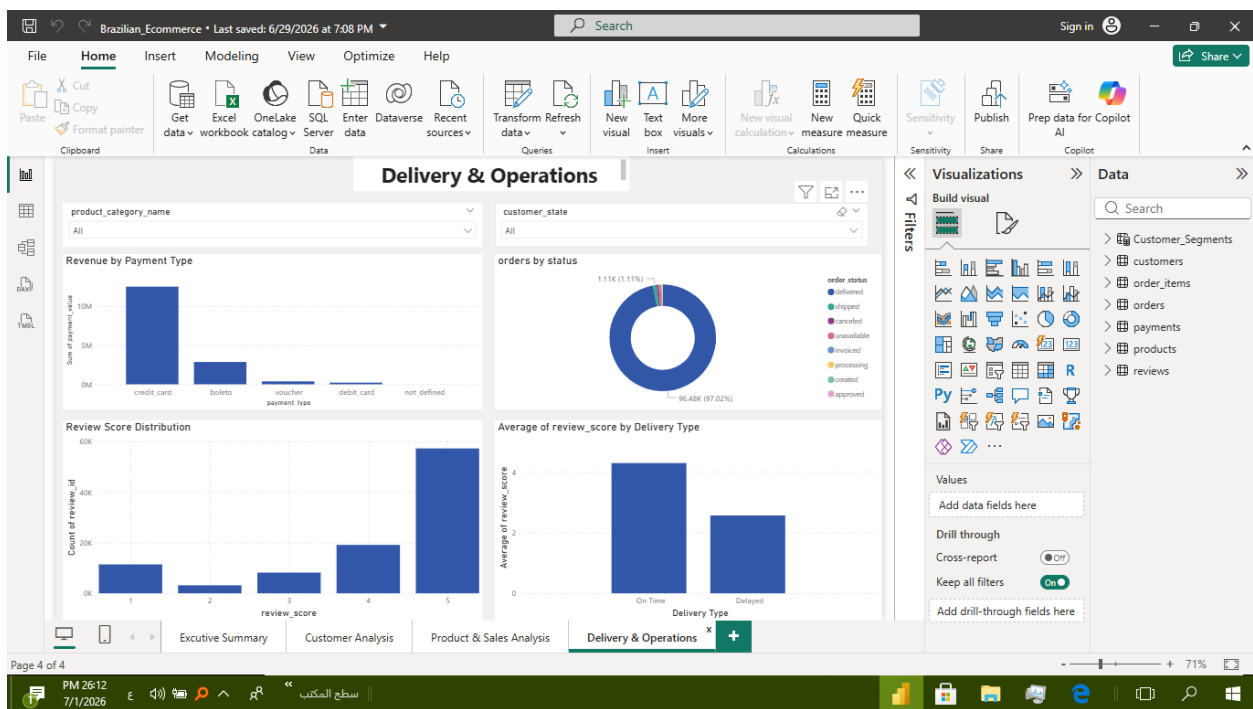
Customer Analysis



Description

This dashboard focuses on customer purchasing behavior, spending patterns, customer loyalty, and geographical distribution. It highlights top customers, customer segmentation, repeat customer metrics, and regional order distribution to support customer-focused business strategies.

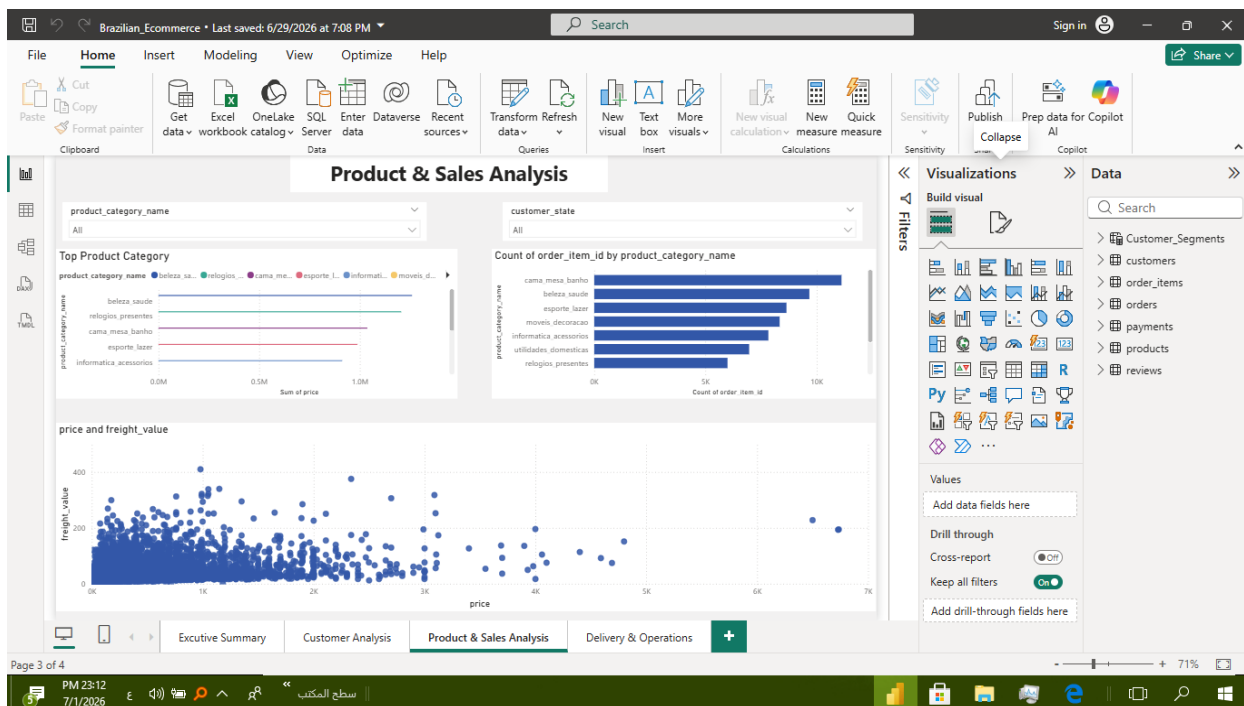
Delivery & Operations Analysis



Description

This dashboard evaluates operational performance by monitoring delivery times, delayed deliveries, customer review scores, and order status distribution. These visualizations help assess logistics efficiency and identify factors affecting customer satisfaction.

Product & Sales Analysis



Description

This dashboard analyzes product performance and sales patterns across different product categories. It highlights the top-performing product categories, the most frequently ordered categories, and the relationship between product price and freight cost. These visualizations help identify high-performing products, evaluate pricing patterns, and support product management and sales strategies.

7. Key Business Insights

The analysis produced several important business insights that summarize the overall performance of the Brazilian e-commerce platform:

- The platform generated over **15 million** in total revenue, demonstrating strong sales performance.
- Sales activity fluctuated throughout the year, indicating seasonal purchasing behavior.
- A small number of product categories generated the majority of total revenue.
- High-value customers contributed a significant portion of overall sales.
- Repeat customers spent more on average than one-time customers, highlighting the importance of customer retention.
- Customer loyalty remained relatively low, suggesting opportunities to improve repeat purchase rates.
- Certain cities and states generated considerably higher order volumes than others.
- Delivery delays negatively affected customer review scores, emphasizing the importance of logistics performance.
- Most orders were successfully delivered, indicating efficient order fulfillment operations.
- Freight costs generally increased with product prices, reflecting the relationship between product value and shipping expenses.

8. Business Recommendations

Based on the analysis, the following recommendations are proposed to improve business performance:

- Strengthen customer loyalty programs to increase repeat purchases.
- Focus marketing campaigns on high-value customers and high-performing regions.
- Increase investment in the best-performing product categories.
- Improve delivery performance to reduce delays and enhance customer satisfaction.
- Optimize freight costs for categories with high shipping expenses.
- Prepare inventory according to seasonal demand patterns.
- Monitor key business KPIs regularly using interactive Power BI dashboards.
- Continue using data-driven decision-making to support future business growth.

9. Conclusion

This project demonstrated how SQL and Power BI can be combined to transform raw business data into meaningful insights.

Using SQL, business questions were answered through data extraction, aggregation, and analytical queries. Power BI was then used to convert these findings into interactive dashboards that support business monitoring and decision-making.

The project provided valuable insights into sales performance, customer behavior, logistics efficiency, and financial trends while demonstrating practical skills in data analysis, business intelligence, and data visualization.

Overall, this project reflects an end-to-end business analytics workflow, from data exploration and SQL analysis to dashboard development and business reporting.